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/ **Practice Exam - 1**

Correct
Answer

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Practice Exam - 1

Question 1 of 117



A 32-year-old female status post orthotopic heart transplantation 6 years prior for non-ischemic cardiomyopathy presents to your office for preconception counseling. Her post transplant course has been unremarkable with no episodes of rejection or infection. Donor specific antibodies are negative. She has developed mild hypotension since transplant which is successfully managed with Norvasc 2.5 mg daily. She is maintained on tacrolimus and mycophenolate for immunosuppression. Her regimen otherwise consists of aspirin 81 mg, pravastatin 40 mg, vitamin C and E.

You plan to refer her to maternal-fetal medicine for more detailed preconception counseling and to establish a multidisciplinary team for her management before conception, during pregnancy and in the postpartum period. She asks you about management of her immunosuppressive regimen from preconception to postpartum.

Which of the following statements is correct in counseling her?

- ☒ **A.** Transition from tacrolimus to sirolimus
- ☐ **B.** Discontinuation of mycophenolate mofetil before planned conception
- ☐ **C.** Discontinuation of tacrolimus from first trimester until completion of breast feeding
- ☐ **D.** Discontinuation of tacrolimus during the first trimester only

Commentary

References

Testing Point: The Board-certified AHFTC specialist should know how immunosuppression must be adjusted when a heart transplant female recipient plans a pregnancy.

Answer: B. Discontinuation of mycophenolate mofetil before planned conception.

She will need to discontinue mycophenolate mofetil 6-12 weeks before planned conception due to severe teratogenic effects.

Mycophenolate mofetil was categorized as category D by the FDA under the old classification system and currently has a black box warning for embryofetal toxicity. Mycophenolate has known teratogenicity and can cause structural malformations including anophthalmia, adenopathy, hydrocephaly and fetal death.

As such it is recommended that mycophenolate be discontinued 6 to 12 weeks prior to conception or at minimum immediately after unplanned pregnancy is discovered. This is detailed further in a 2018 ISHLT consensus report. This is also given a class I evidence level C indication in the 2010 ISHLT guidelines.

This was further highlighted in the recent publication in JHLT highlighting 157 pregnancies in 91 female heart transplant patients in North America. Live births occurred in 69% of pregnancies, there were no neonatal deaths, a rejection rate of 9.1% was reported, and other pregnancy complications including preeclampsia were seen in 23% of patients and infections in 14% of patients. Other series have reported higher rejection rates up to 19%.

This study also demonstrated that of pregnancies exposed to mycophenolate, 63% had miscarriages and 23% of live births suffered birth defects (compared to miscarriage percentage of 17% and a birth defect percentage of 6% when not exposed to mycophenolate, and miscarriage rate of 15-20% and birth defect rate of 2-4% in all clinically recognized pregnancies). The birth defects in this study included duodenal atresia, laryngomalacia and facial defects.

In general, it is recommended to delay planned pregnancies until after the first year after heart transplantation due to the high rejection risk in the first year. Pregnancy is typically avoided in heart transplant patients with LV dysfunction, significant CAD, significant and/or uncontrolled comorbidities and in those who have donor specific antibodies and at higher risk of rejection. Ideally the transplant recipient should have no rejections in the prior year, "stable graft function", no recent infections and stable immunosuppressive dosing.

Answer A is incorrect as the ISHLT consensus report also recommends discontinuation of proliferation signal inhibitors (PSI) 6 to 12 weeks prior to conception. Data on PSI during pregnancy is very limited however animal studies show teratogenicity including fetal demise, structural malformations and growth retardation. Prior FDA classification was category C. The current risk assessment is that PSIs are associated with risk to the fetus which is unquantified at present.

Answers C and D are incorrect as tacrolimus has not been shown to be teratogenic in humans and tacrolimus use during pregnancy is considered "acceptable" as described by the 2018 ISHLT consensus statement. There is a concern for kidney dysfunction and hyperkalemia in the newborn who was exposed to tacrolimus during late pregnancy. Breast-feeding is also considered acceptable with tacrolimus. Tacrolimus was previously categorized by the FDA as category C.

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